Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 20 epochs with a batch size of 128. Each experiment was repeated 5 times with different random seeds.

2 Results

Figure 1 shows the training loss.

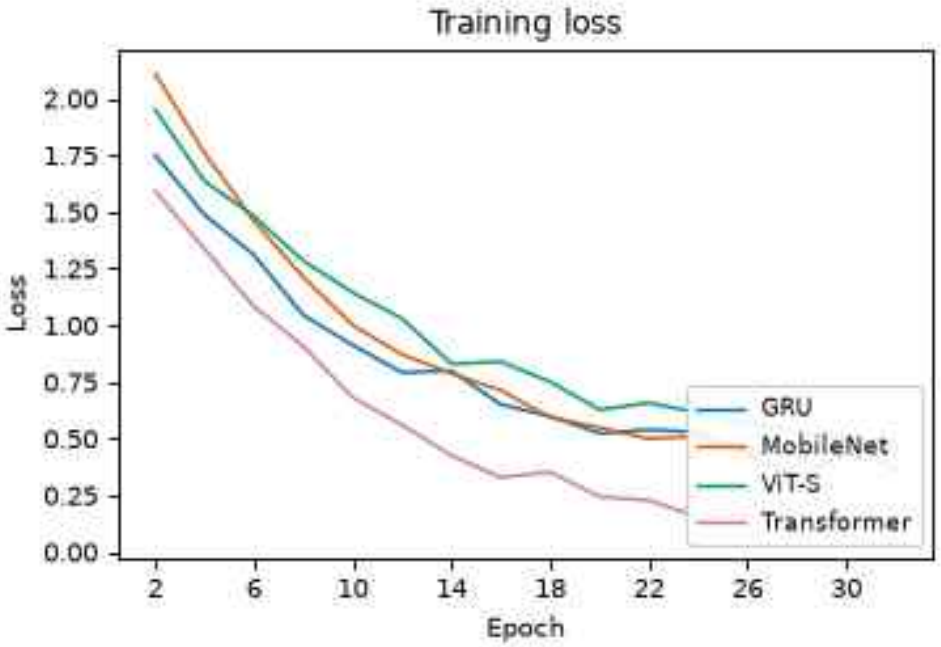


Figure 1: Training loss for the models in this experiment.

ViT-S has a loss of over 0.8 at epoch 14 (Figure 1). As shown in Figure 1, ViT-S has a loss of 0.434 at epoch 22. At epoch 4, the loss of GRU is 1.486 (Figure 1). At epoch 4, the loss of Transformer is 1.782.

Figure 2 shows the f1 score of EfficientNet by dataset.

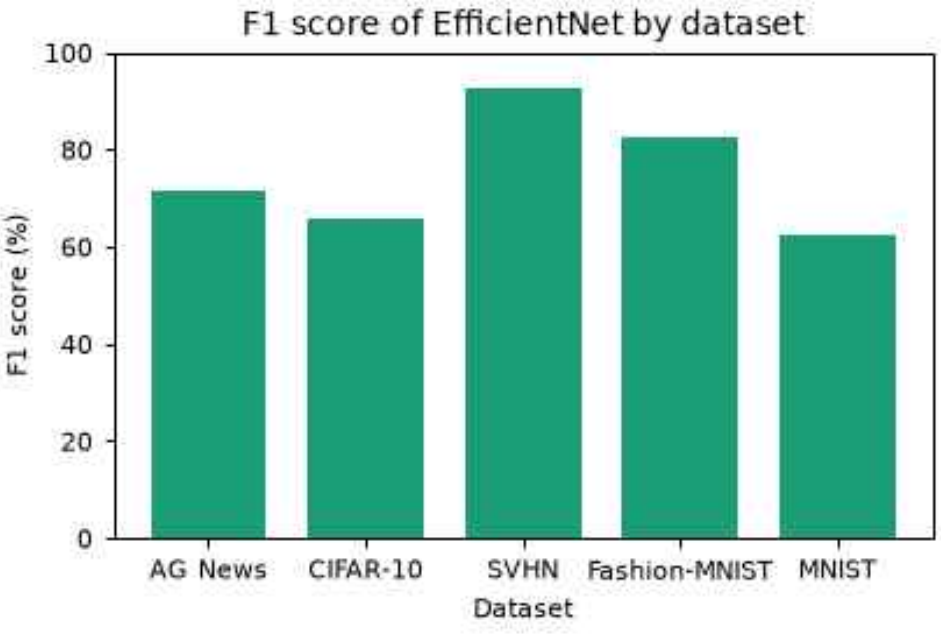


Figure 2: F1 score of EfficientNet by dataset for the models in this experiment.

As shown in Figure 2, EfficientNet scores 40.1 points higher on SVHN than on MNIST. EfficientNet scores 25.9 points higher on AG News than on CIFAR-10. EfficientNet reaches 75.6% F1 score on CIFAR-10. EfficientNet scores 80.1 points higher on SVHN than on MNIST.

Figure 3 shows the test accuracy during training.

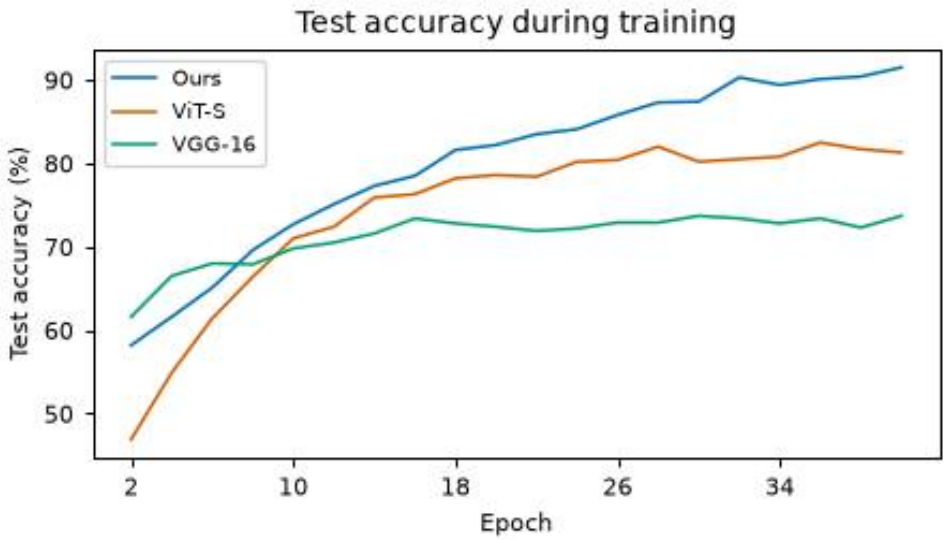


Figure 3: Test accuracy during training for the models in this experiment.

Ours outperforms ViT-S by 28.4 percentage points at epoch 24. At epoch 18, ViT-S achieves 78.2% test accuracy.

3 Discussion

On Fashion-MNIST, Random Forest obtains an accuracy of 92.7%. Overall, the results are in line with what we expected from the literature.